

	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1

SCOPE OF WORKS:

Epiroc ST18



	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1

Overview

1. Project Background and Description

This document will define the agreed equipment, functionality and responsibilities of Strata and Newcrest Cadia to ensure supply and installation of the HazardAvert PDS performs as expected.

2. Purpose of HazardAvert PDS

- To provide a warning to personnel entering the “No Go” zones around the machine/vehicle
- To warn the machine operator of personnel entering the “No Go” zones around the machine/vehicle
- To activate the HazardAvert System when the machine is moving and de-activate the system when the machine is not moving

3. Installation Scope

Equipment hardware requirements

The Epiroc ST18 will be fitted with a dual Generator proximity system consisting of the following equipment:

- 1 x HA-SHR-IM-01: 12/24 Volt Controller Box.
- 2 x HA-SHR-PM-01: Proximity Module (generator).
- 1 x HA-SHR-DSPLAY-01: 4.3” CANbus display
- 1 x Silent Zone Loop
- Cabling and consumables for all the above equipment

Excluded are the brackets required for the Generator, which will be provided by Newcrest Cadia

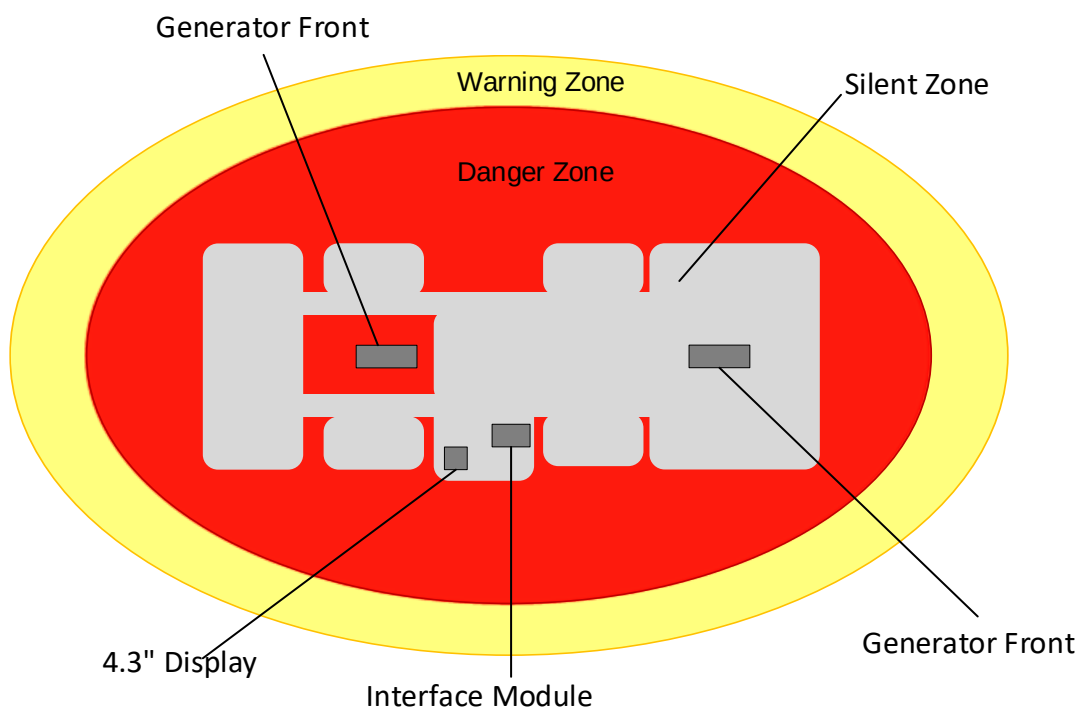


Figure 1: General arrangement of HA Equipment Layout and zones

	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1

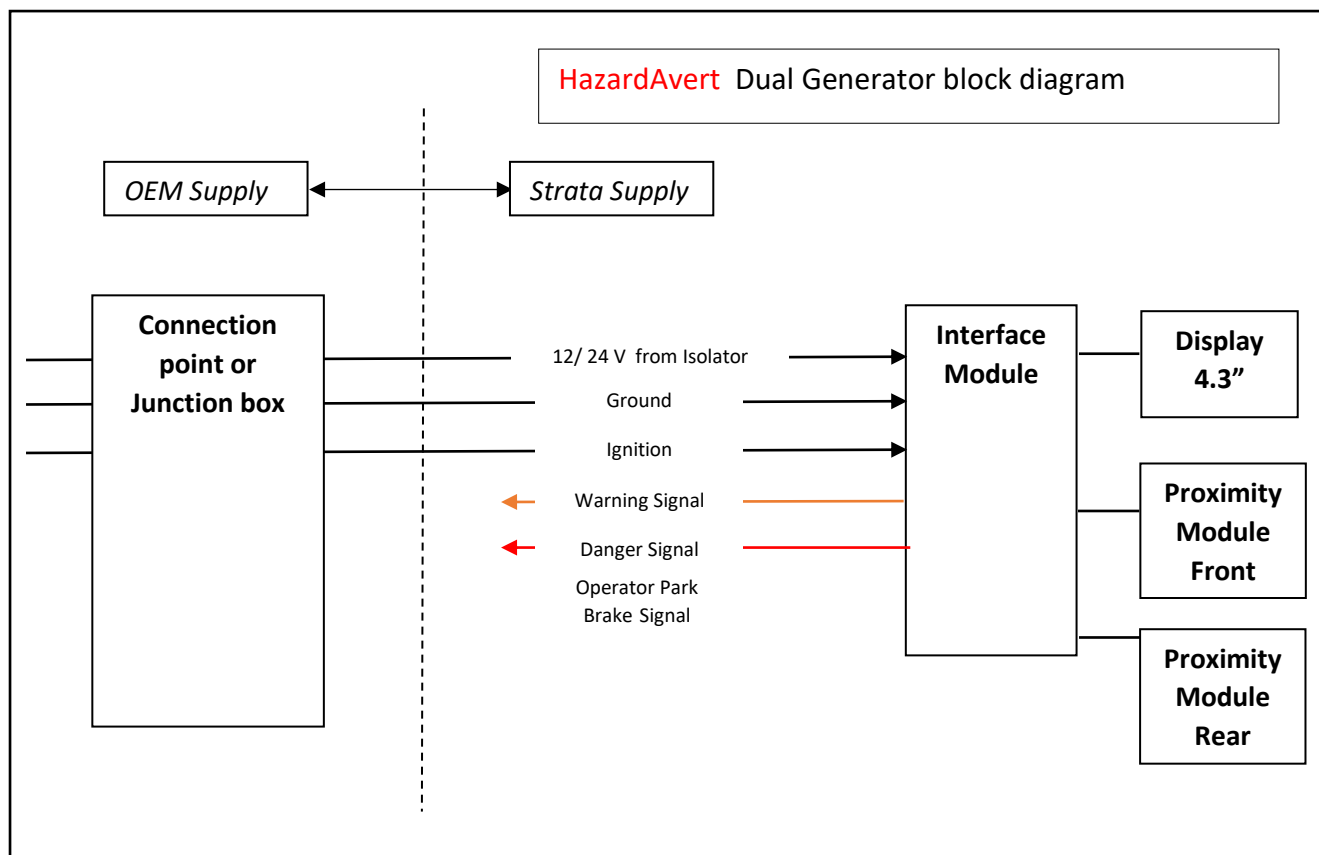


Figure 2: Hardware and connection layout

Definition of Danger Zone

HazardAvert™ establishes an electromagnetic marker field or zone around machinery using a device called the generator. One or more generators may be fitted onto machinery and create fields designated as the ‘WARNING ZONE’ and ‘DANGER ZONE’. This marker field can encompass the entire machine and its turning radius, or be specific to certain areas. Personnel wear a Personal Alarm Device (PAD) which detects and measures the magnetic marker field to determine their proximity to the machine, and alert them of possible danger to their safety. The warning buzzer emits audible and visual alarms. The generator also incorporates a vehicle alarm device (VAD) which detects the electromagnetic fields of other machinery to help avoid vehicle collision.



Figure 1. Generator Magnetic Field Detection

	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1

Definition of Activate Fields Signal

The magnetic marker field described above is only required when the machine's park brake is released. HazardAvert is not required to alarm when the machine is considered to be safe.

The activate fields signal are provided to the HazardAvert system when:

- An input from the Operator Park Brake provides a +12/24V signal to the HA Controller unit when the Operator Park Brake is in de-activated and the ignition is on.

Note: A risk assessment should be completed to ensure the above definition of a safe machine state is adequate for the customer's specific site requirements.

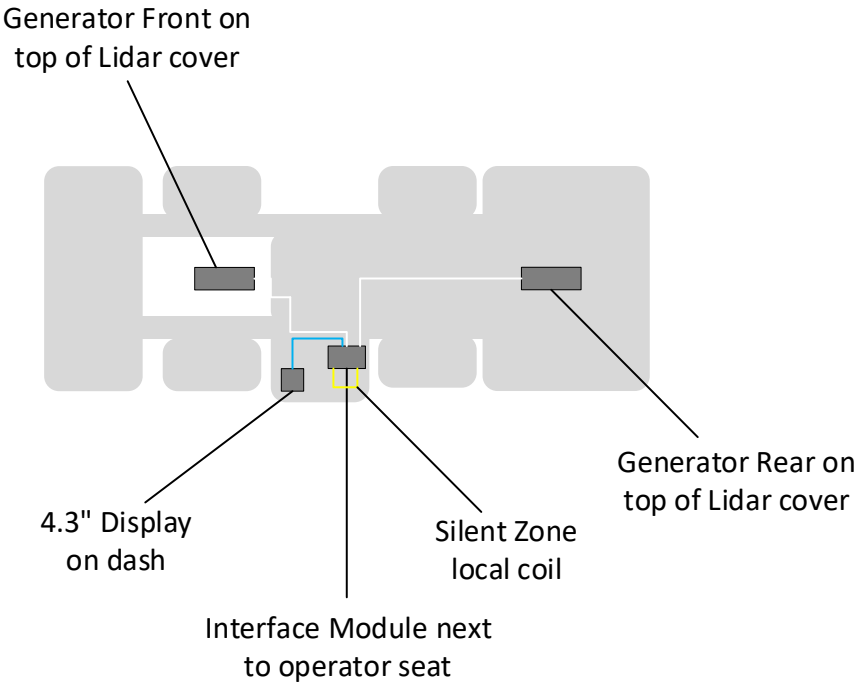
Installation notes

Caution:

- Consult fleet supervisor prior to drilling into vehicle/machine
- No modifications are to be made on ROPs (Roll over Protection structure) or on FOPs (Falling object Protective structures)
- Follow all existing OEM cable routes. Do not cable tie onto or follow the fire suppression system on the machine. Do not route or secure to hydraulic hoses. Keep cable routes away from potential impact, rub points and moving parts.
- Magnets with ladders are acceptable where needed for cable routes
- Apply Densyl tape to all connections outside of the cab
- All external cabling must be run in corrugated conduit (no split conduit). Adhesive heat shrink to be applied on the Generator cables to prevent water ingress into the conduit, at both conduit start and end points.
- All cables installed internal (inside cab) must run inside finger trap / guard flex
- Cables must be labelled as per MDG15 standard
- Ensure all bracketry is prepared before installation
- Do not impeded driver's vision with placement of screen

	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1

Epiroc ST18



Location of Interface Module (IM)

- Mount the interface module next to the operator’s seat with magnets

Note: Test to ensure no cables are pinched during seat adjustments. The Deutsch plugs must rear.



	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1



Location of Proximity Module (PM) Front

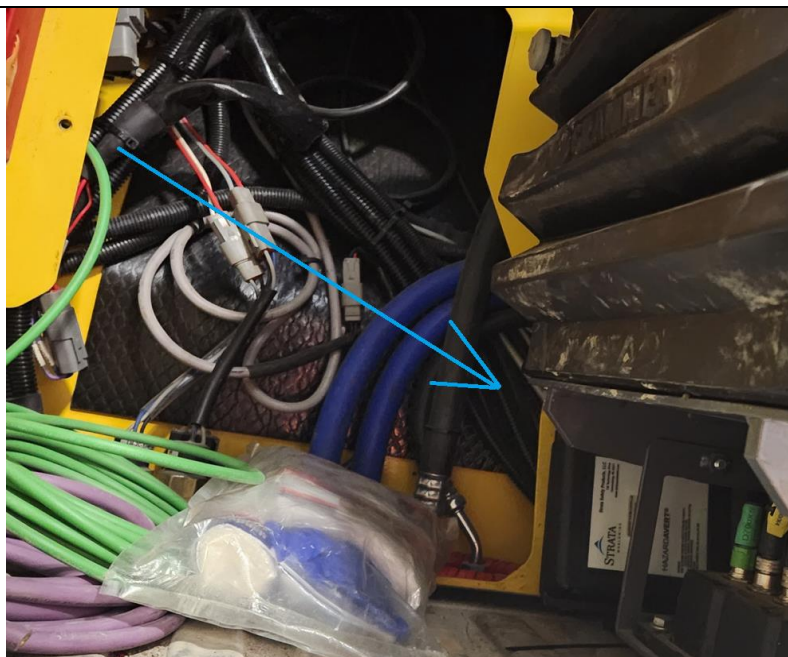
- Mount the Proximity Module on the front Lidar cover. Cover must have 4 x M6 holes drilled and tapped.
- Proximity Module to be mounted on rubber standoffs, with Amphenol connector facing the rear. Cable to be inserted in corrugated conduits (non-split) and waterproofed by securing adhesive heat shrink where conduit starts and ends.
- Cable to follow existing cable looms
- Make sacrificial harness at articulation point (make a spare and label appropriately)
- Continue cable run by following existing cable looms into cab and behind covers to IM next to the seat



	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1



	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1



Location of Proximity Module (PM) Rear

- Mount the Proximity Module on the rear Lidar cover. Cover must have 4 x M6 holes drilled and tapped.
- Proximity Module to be mounted on rubber standoffs, with Amphenol connector end facing the front. Cable to be inserted in corrugated conduits (non-split) and waterproofed by securing adhesive heat shrink where conduit starts and ends
- Cable to follow existing cable looms underneath engine panel lids into cab



	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1

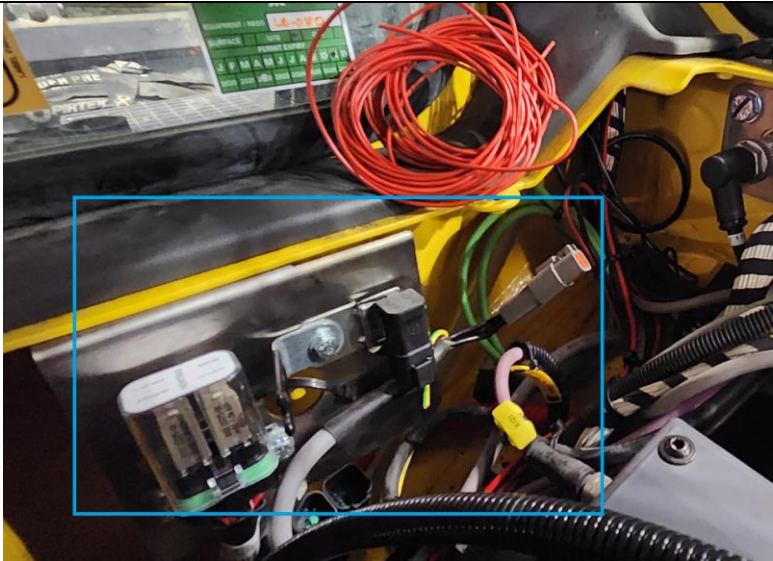


Location of the 4.3" CANbus display

- Mount the display on right hand side of door in the corner next to the big screen
- Drill a 20mm hole behind screen bracket and insert compression gland
- Run cable through compression gland and underneath dash towards the right-hand side, then down to the floor and towards the seat



	Epiroc ST18 Interface Agreement	Effective Date /2023	Doc Number SPA-ENG-SOW-230003
		Doc Type Scope of Works	Revision 1

<u>Location of Silent Zone Cable</u> Use short SZ coil secured next to IM	
<u>Power connection</u> - Get DC supply from Newtrax old supply behind panel inside cab. Make sure power is triggered from ignition (IGN) using Deutch plug of Newtrax: Ignition +24V, Supply +24VDC, ground 0V - Run power cable and IGN following existing OEM cables - Follow cables in cab and run safely to the IM next to the operator's seat	
<u>Park brake Sensor</u> Connect to GND of park brake sensor located at A10 bulk head switch S138. Pin 24 - 0V closed when brakes applied	